

## 0.1 Riemann-Stieltjes Integral

**Definition 0.1.1.** A finite subset  $P \subset [a, b]$  containing  $a, b$  is called a *Partition* of  $[a, b]$ . Conventionally, denote  $P = \{x_0, x_1, \dots, x_n\}$  where  $a = x_0 < x_1 < \dots < x_n = b$ . The set of Partitions of  $[a, b]$  is denoted as  $\mathcal{P}[a, b]$ .

**Definition 0.1.2.** Let  $f: [a, b] \rightarrow \mathbb{R}$  be a bounded map, and  $\alpha: [a, b] \rightarrow \mathbb{R}$  be a monotone-increasing. Let a partition  $P \in \mathcal{P}[a, b]$  be given, and denote  $P = \{x_0, x_1, \dots, x_n\}$ . For each  $1 \leq i \leq n$ , define

$$\Delta\alpha_i \stackrel{\text{def}}{=} \alpha(x_i) - \alpha(x_{i-1}) \underset{\alpha \text{ mono.}}{\geq} 0$$

and

$$M_i \stackrel{\text{def}}{=} \sup\{f(t) \mid x_{i-1} \leq t \leq x_i\} \text{ and } m_i \stackrel{\text{def}}{=} \inf\{f(t) \mid x_{i-1} \leq t \leq x_i\}$$

And, define *Upper Sum* and *Lower Sum* as:

$$U(f, P, \alpha) \stackrel{\text{def}}{=} \sum_{i=1}^n M_i \cdot \Delta\alpha_i \text{ and } L(f, P, \alpha) \stackrel{\text{def}}{=} \sum_{i=1}^n m_i \cdot \Delta\alpha_i$$

Now, define *Upper integral* and *Lower integral* as:

$$\overline{\int_a^b} f d\alpha \stackrel{\text{def}}{=} \inf_{P \in \mathcal{P}[a, b]} U(f, P, \alpha) \text{ and } \underline{\int_a^b} f d\alpha \stackrel{\text{def}}{=} \sup_{P \in \mathcal{P}[a, b]} L(f, P, \alpha)$$

We say that  $f$  is *Riemann-Stieltjes Integrable* with respect to  $\alpha$  if:

$$\overline{\int_a^b} f d\alpha = \underline{\int_a^b} f d\alpha$$

The set of Riemann-Stieltjes Integrable functions with respect to  $\alpha$  as  $\mathcal{R}(\alpha)$ .

**Lemma 0.1.3.** Let  $f: [a, b] \rightarrow \mathbb{R}$  be a bounded map, and  $\alpha: [a, b] \rightarrow \mathbb{R}$  be a monotone-increasing. For any partitions  $P, Q \subset [a, b]$ , and a refinement  $P^* \supset P$ ,

$$L(f, P, \alpha) \leq L(f, P^*, \alpha) \leq \underline{\int_a^b} f d\alpha \leq \overline{\int_a^b} f d\alpha \leq U(f, P^*, \alpha) \leq U(f, P, \alpha)$$

and

$$L(f, P, \alpha) \leq U(f, Q, \alpha)$$

Proof of Lemma 0.1.3.

Let  $P^* \supset P$  be a refinement of  $P$ . Since partition contains only finite elements, WLOG, assume that  $P^* = P \cup \{s\}$  s.t.  $s \notin P = \{x_0, \dots, x_n\}$

Then, for some  $1 \leq j \leq n$   $x_{j-1} < s < x_j$ . Now,

$$\begin{aligned} U(f, P, \alpha) - U(f, P^*, \alpha) &= \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(x_j) - \alpha(s)] + \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(s) - \alpha(x_{j-1})] \\ &\quad - \sup_{t \in [x_{j-1}, s]} f(t) \cdot [\alpha(x_j) - \alpha(s)] - \sup_{t \in [s, x_j]} f(t) \cdot [\alpha(s) - \alpha(x_{j-1})] \\ &\geq 0. \end{aligned}$$

### 0.1.1 Properties of Riemann-Stieltjes Integral

**Theorem 0.1.4.** Let  $f : [a, b] \rightarrow \mathbb{R}$  be a bounded map, and  $\alpha : [a, b] \rightarrow \mathbb{R}$  be monotone-increasing. TFAE:

1.  $f$  is Riemann-Stieltjes Integrable with respect to  $\alpha$  on  $[a, b]$ .
2. For any  $\varepsilon > 0$ , there exists a Partition  $P \in \mathcal{P}[a, b]$  such that  $U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon$ .
3. For some  $A \in \mathbb{R}$ , for any  $\varepsilon > 0$ , there exists  $P_0 \in \mathcal{P}[a, b]$  such that

$$\forall P \in \mathcal{P}[a, b], P \supseteq P_0 \implies |S(f, P, \alpha) - A| < \varepsilon$$

where Riemann-Stieltjes Sum  $S(f, P, \alpha) \stackrel{\text{def}}{=} \sum_{i=1}^n f(t_i)[\alpha(x_i) - \alpha(x_{i-1})]$ , for some  $t_i \in [x_{i-1}, x_i]$ .

**Lemma 0.1.5.** Let  $f : [a, b] \rightarrow \mathbb{R}$  be a bounded map, and  $\alpha : [a, b] \rightarrow \mathbb{R}$  be a monotone-increasing.

If  $f, g \in \mathcal{R}(\alpha)$  and  $c \in \mathbb{R}$ , then  $f + g, cf \in \mathcal{R}(\alpha)$  with

$$\begin{aligned}\int_a^b (f + g) d\alpha &= \int_a^b f d\alpha + \int_a^b g d\alpha \\ \int_a^b (cf) d\alpha &= c \int_a^b f d\alpha\end{aligned}$$

**Lemma 0.1.6.** Let  $f : [a, b] \rightarrow \mathbb{R}$  be a bounded map, and  $\alpha : [a, b] \rightarrow \mathbb{R}$  be a monotone-increasing, and  $a < c < b$ .

$f$  is Stieltjes Integrable on  $[a, b]$  if and only if  $f$  is Stieltjes Integrable on  $[a, c]$  and  $[c, b]$

In this case, the following equality holds:

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha$$

**Lemma 0.1.7.** Let  $f : [a, b] \rightarrow \mathbb{R}$  be a bounded map, and  $\alpha, \beta : [a, b] \rightarrow \mathbb{R}$  be monotone-increasing.

If  $f \in \mathcal{R}(\alpha)$ ,  $f \in \mathcal{R}(\beta)$  and  $c > 0$ , then  $f \in \mathcal{R}(\alpha + \beta)$  and  $f \in \mathcal{R}(c\alpha)$  and  $|f| \in \mathcal{R}(\alpha)$  with

$$\begin{aligned}\int_a^b f d(\alpha + \beta) &= \int_a^b f d\alpha + \int_a^b f d\beta \\ \int_a^b f d(c\alpha) &= c \int_a^b f d\alpha \\ \left| \int_a^b f d\alpha \right| &\leq \int_a^b |f| d\alpha\end{aligned}$$

Proof of Thm 0.1.4

1)  $\Rightarrow$  2). Since  $f \in \mathcal{R}(a)$ , given  $\varepsilon > 0$ , there exists a partition  $P \in \mathcal{P}[a, b]$  s.t.

$$\int_a^b f \, d\alpha \leq U(f, P, \alpha) < \int_a^b f \, d\alpha + \frac{\varepsilon}{2} \quad \text{and} \quad \int_a^b f \, d\alpha \geq L(f, P, \alpha) > \int_a^b f \, d\alpha - \frac{\varepsilon}{2}$$

Ask the next step to a toddler on the street.

2)  $\Rightarrow$  1). It is clear, because for any partition  $P \in \mathcal{P}[a, b]$ ,

$$\int_a^b f \, d\alpha - \int_a^b f \, d\alpha \leq U(f, P, \alpha) - L(f, P, \alpha)$$

To show 1)  $\Leftrightarrow$  3), we prove a statement: If 2) holds, then for any  $1 \leq i \leq n$ ,

$$\text{for arbitrary } s_i, t_i \in [x_{i-1}, x_i], \quad \sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i < \varepsilon.$$

$$\text{Since } |f(s_i) - f(t_i)| \leq M_i - m_i,$$

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i \leq U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

1)  $\Rightarrow$  3). Suppose  $f \in \mathcal{R}(a)$ . Then, by 2), there exists a partition  $P_0 \in \mathcal{P}[a, b]$  s.t.

$$U(f, P_0, \alpha) - L(f, P_0, \alpha) < \varepsilon$$

Now, for any refinement  $P > P_0$ ,

$$U(f, P, \alpha) - L(f, P, \alpha) \leq U(f, P_0, \alpha) - L(f, P_0, \alpha) < \varepsilon$$

Clearly, regardless of choice of  $t_i$ ,

$$L(f, P, \alpha) \leq S(f, P, \alpha) \leq U(f, P, \alpha)$$

$$\text{and} \quad U(f, P, \alpha) \geq \int f \, d\alpha > L(f, P, \alpha)$$

$$\text{Therefore, } |S(f, P, \alpha) - \int f \, d\alpha| \leq U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

3)  $\Rightarrow$  1) is trivial.

Proof of Lemma 0.1.5.

Let  $f, g \in \mathcal{R}(a)$  be given. Given  $\epsilon > 0$ , there exists a partition  $P \in \mathcal{P}[a, b]$  such that

$$U(f, P, \alpha) - L(f, P, \alpha) < \frac{\epsilon}{2} \quad \text{and} \quad U(g, P, \alpha) - L(g, P, \alpha) < \frac{\epsilon}{2}$$

(Technically, take  $P_1, P_2 \in \mathcal{P}[a, b]$  for  $f, g$  respectively, and put  $P = P_1 \cup P_2$ ). Now,

$$U(f+g, P, \alpha) - L(f+g, P, \alpha) \leq U(f, P, \alpha) + U(g, P, \alpha) - L(f, P, \alpha) - L(g, P, \alpha) < \epsilon.$$

And the equality is obtained by: given  $\epsilon > 0$ , there exists a partition  $P \in \mathcal{P}[a, b]$  s.t.

$$\begin{aligned} \int_a^b f \, d\alpha + \int_a^b g \, d\alpha - \epsilon &\leq L(f, P, \alpha) + L(g, P, \alpha) \leq L(f+g, P, \alpha) \\ &\leq \int_a^b f+g \, d\alpha \\ &\leq U(f+g, P, \alpha) \leq U(f, P, \alpha) + U(g, P, \alpha) \leq \int_a^b f \, d\alpha + \int_a^b g \, d\alpha + \epsilon \end{aligned}$$

Proof of Lemma 0.1.6.

Suppose that  $f \in \mathcal{R}(a)$  on  $[a, b]$ . Given  $\epsilon > 0$ , there exists  $P \in \mathcal{P}[a, b]$  s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \epsilon.$$

Put  $P^* = P \cup \{c\}$ . Then, for  $P_1 = P^* \cap [a, c]$  and  $P_2 = P^* \cap [c, b]$ . Now,

$$\begin{aligned} \int_a^c f \, d\alpha - \int_a^c f \, d\alpha + \int_c^b f \, d\alpha - \int_c^b f \, d\alpha &\leq U(f, P_1, \alpha) - L(f, P_1, \alpha) + U(f, P_2, \alpha) - L(f, P_2, \alpha) \\ &= U(f, P^*, \alpha) - L(f, P^*, \alpha) < \epsilon. \end{aligned}$$

Similarly, the converse holds.

Thm. Let  $f: [a, b] \rightarrow \mathbb{R}$  be a bounded map and continuous except finitely many points. Then  $f$  is Riemann integrable.

Proof. Define  $P_n = [a, b] \cap \left\{ \frac{k}{2^n} \mid k \in \mathbb{Z} \right\}$ . Let  $A$  be a set of discontinuous points of  $f$ .

Put  $N$  be a number of elements of  $A$ . Now, for any  $n \in \mathbb{N}$ , there is at most  $2N$  for  $i \in \mathbb{N}$

$$\text{s.t. } A \cap [x_{i-1}, x_i] \neq \emptyset$$

where  $x_i \in P_n$ . Given  $\varepsilon > 0$ , take  $n \in \mathbb{N}$  s.t.  $2 \cdot \sup |f| \cdot 2N \cdot \frac{1}{2^n} < \frac{\varepsilon}{2}$ .

Meanwhile,  $f$  is continuous on the set

$$C = \bigcup_{A \cap [x_{i-1}, x_i] = \emptyset} [x_{i-1}, x_i]$$

Therefore there is a partition  $P'$  on  $C$  s.t.  $U(f, P') - L(f, P') < \frac{\varepsilon}{2}$ .

Finally, for  $P = P_n \cup P'$ ,

$$U(f, P) - L(f, P) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} < \varepsilon. \quad \square$$

### 0.1.2 Riemann-Stieltjes Integrable

**Theorem 0.1.8.** If  $f: [a, b] \rightarrow \mathbb{R}$  is continuous, then  $f \in \mathcal{R}(\alpha)$  for any monotone increasing  $\alpha: [a, b] \rightarrow \mathbb{R}$ .

**Theorem 0.1.9.** If  $f: [a, b] \rightarrow \mathbb{R}$  is monotone and  $\alpha: [a, b] \rightarrow \mathbb{R}$  is monotone increasing continuous, then  $f \in \mathcal{R}(\alpha)$ .

**Theorem 0.1.10.** If  $f: [a, b] \rightarrow \mathbb{R}$  is map with  $m \leq f(t) \leq M$  for all  $t \in [a, b]$  and  $\phi$  is continuous on  $[m, M]$ , then  $\phi \circ f$  is Riemann-Stieltjes integrable with respect to  $\alpha$  on  $[a, b]$

**Theorem 0.1.11.** If  $\alpha: [a, b] \rightarrow \mathbb{R}$  is continuously differentiable monotone increasing, then

$$f \in \mathcal{R}(\alpha) \text{ if and only if } f\alpha' \in \mathcal{R}(x)$$

In this case,

$$\int_a^b f(x) d\alpha(x) = \int_a^b f(x) \alpha'(x) dx$$

*Proof of Thm 0.1.8.*

Given  $\epsilon > 0$ , take  $\eta > 0$  s.t.  $[\alpha(b) - \alpha(a)] \cdot \eta < \epsilon$ .

Since  $f$  is continuous on the compact set  $[a, b]$ , it is continuous uniformly.

Thus, there exists  $\delta > 0$  s.t.

$$\forall x, y \in [a, b], |x - y| < \delta \Rightarrow |f(x) - f(y)| < \eta.$$

Now, let  $P \in \mathcal{P}[a, b]$  be a partition such that for all  $i$ ,  $\Delta x_i < \delta$  ( $P = \{x_0, \dots, x_n\}$ )

Then, for all  $i$ ,  $M_i - m_i \leq \eta$ . Now,

$$U(f, P, \alpha) - L(f, P, \alpha) = \sum_{i=1}^n (M_i - m_i) \cdot \Delta \alpha_i \leq \sum_{i=1}^n \eta \cdot \Delta \alpha_i = \eta \cdot \sum_{i=1}^n \Delta \alpha_i = \eta \cdot [\alpha(b) - \alpha(a)] < \epsilon.$$

Proof of Thm 0.1.9. Alternatively, since monotone function can have <sup>only</sup> countably simple discontinuity, if no such shared discontinuities with  $f$  and  $\alpha$ , then countably discontinuous  $f$  is R.S. integrable. But, in this, prove by elementary, using properties of monotone.

Since  $\alpha$  is continuous, for each  $n \in \mathbb{N}$ , there exists a partition  $P_n \in \mathcal{P}[a, b]$ ,

$$\Delta \alpha_i = \frac{\alpha(b) - \alpha(a)}{n} \quad \text{for all } i$$

(More specifically, let  $n \in \mathbb{N}$  be fixed. By the Intermediate Value Theorem, there exists  $x_1 \in [a, b]$  s.t.  $\alpha(x_1) = \alpha(a) + \frac{\alpha(b) - \alpha(a)}{n}$ . Inductively,

there exists  $x_2 \in [x_1, b]$  s.t.  $\alpha(x_2) = \alpha(a) + 2 \cdot \frac{\alpha(b) - \alpha(a)}{n}$

$$\vdots$$

$$x_{n-1} \in [x_{n-2}, b] \text{ s.t. } \alpha(x_{n-1}) = \alpha(a) + (n-1) \cdot \frac{\alpha(b) - \alpha(a)}{n}$$

And take  $x_n = b$ . Essentially, the Existence such partition is obtained by I.V.P.

But, Monotone can have only first-kind discontinuity and

IVP implies it can not have first-kind discontinuity, thus Monotone I.V.T  $\Rightarrow$  Continuity.

Given  $\epsilon > 0$ , <sup>wlog,</sup> Assume  $f$  is monotone increasing. Take  $n \in \mathbb{N}$  s.t.  $\frac{[\alpha(b) - \alpha(a)] \cdot [f(b) - f(a)]}{n} < \epsilon$ .

$$U(f, P_n, \alpha) - L(f, P_n, \alpha) = \sum_{i=1}^n [M_i - m_i] \cdot \Delta \alpha_i = \sum_{i=1}^n [f(x_i) - f(x_{i-1})] \cdot \Delta \alpha_i = \Delta \alpha \cdot \sum_{i=1}^n [f(x_i) - f(x_{i-1})]$$

$$= \Delta \alpha \cdot [f(b) - f(a)] < \epsilon. \quad \square$$

Proof of Thm. 0.1.10.

Since  $\phi$  is continuous on the compact set  $[m, M]$ , it is continuous uniformly.

Given  $\epsilon > 0$ , there exists  $0 < \delta < \epsilon$  s.t.  $|s - t| < \delta \Rightarrow |\phi(s) - \phi(t)| < \epsilon$ .

Since  $f \in \mathcal{R}(\alpha)$  on  $[a, b]$ , there is a partition  $\{x_0, \dots, x_n\}$  on  $[a, b]$  s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \delta^2$$

Let  $M_i = \sup f(t)$  and  $m_i = \inf f(t)$  and  $M_i^* = \sup(\phi \circ f(t))$  and  $m_i^* = \inf(\phi \circ f(t))$  where the sup, inf values are taken in  $t \in [x_{i-1}, x_i]$ ,

Divides the index set  $\{1, \dots, n\}$  as:

$$A = \{i \in \{1, \dots, n\} \mid M_i - m_i < \delta\} \text{ and } B = \{i \in \{1, \dots, n\} \mid M_i - m_i \geq \delta\}.$$

Now, if  $i \in A$ , then  $M_i^* - m_i^* = \sup(\phi \circ f(t)) - \inf(\phi \circ f(t)) \leq \epsilon$

because  $\forall s, t \in [x_{i-1}, x_i]$ ,  $|f(s) - f(t)| \leq M_i - m_i \Rightarrow |\phi(f(s)) - \phi(f(t))| < \epsilon$ .

And, if  $i \in B$ , then  $M_i^* - m_i^* \leq 2 \cdot \sup_{t \in [a, b]} |\phi(t)|$

We obtain that

$$\delta \cdot \sum_{i \in B} \Delta \alpha_i \leq \sum_{i \in B} [M_i - m_i] \cdot \Delta \alpha_i \leq \sum_{i=1}^n [M_i - m_i] \cdot \Delta \alpha_i = U(f, P, \alpha) - L(f, P, \alpha) < \delta^2.$$

That is  $\sum_{i \in B} \Delta \alpha_i < \delta$ . Finally,

$$\begin{aligned} \sum_{i=1}^n [M_i^* - m_i^*] \Delta \alpha_i &= \sum_{i \in A} [M_i^* - m_i^*] \Delta \alpha_i + \sum_{i \in B} [M_i^* - m_i^*] \Delta \alpha_i \\ &\leq \epsilon \cdot \sum_{i \in A} \Delta \alpha_i + 2 \cdot \sup_{t \in [m, M]} |\phi| \cdot \sum_{i \in B} \Delta \alpha_i \\ &\leq \epsilon \cdot [\alpha(b) - \alpha(a)] + 2 \sup |\phi| \cdot \delta \\ &\leq \epsilon \cdot [\alpha(b) - \alpha(a) + 2 \sup |\phi|] \end{aligned}$$



Proof of Thm 0.11.

Suppose that  $f \in \mathcal{R}(a)$ . Given  $\epsilon > 0$ , there is a partition  $P = \{x_0, \dots, x_n\}$  on  $[a, b]$  s.t.  
 $U(f, P) - L(f, P) < \epsilon$ , since  $f'$  is continuous, thus  $f' \in \mathcal{R}$ .

By the Mean-Value Theorem, for each  $1 \leq i \leq n$

$$\Delta f_i = f(x_i) - f(x_{i-1}) = f'(t_i)(x_i - x_{i-1}) = f'(t_i) \cdot \Delta x_i \text{ for some } t_i \in [x_{i-1}, x_i]$$

For any  $s_i \in [x_{i-1}, x_i]$ ,

$$\sum_{i=1}^n |f'(s_i) - f'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n [M_i - m_i] \cdot \Delta x_i < \epsilon.$$

$$\begin{aligned} \text{Now, } & \left| \sum_{i=1}^n f(s_i) \cdot \Delta f_i - \sum_{i=1}^n f(s_i) \cdot f'(s_i) \cdot \Delta x_i \right| \\ &= \left| \sum_{i=1}^n f(s_i) \cdot f'(t_i) \cdot \Delta x_i - \sum_{i=1}^n f(s_i) \cdot f'(s_i) \cdot \Delta x_i \right| \\ &= \left| \sum_{i=1}^n [f'(t_i) - f'(s_i)] \cdot f(s_i) \cdot \Delta x_i \right| \leq \sup |f| \cdot \epsilon \quad (*) \end{aligned}$$

$$\text{Therefore, } \left| \int_a^b f \, dx - \int_a^b f f' \, dx \right| \leq |U(f, P, f) - U(f f', P, x)| \leq \sup |f| \cdot \epsilon$$

Since (\*) holds for any refinement of  $P$ .

### 0.1.3 Riemann-Integral

**Definition 0.1.12.** Let  $\mathcal{P}[a, b]$  be a set of partitions. Define a Norm as:

$$\|\cdot\| : \mathcal{P}[a, b] \rightarrow \mathbb{R} : P \mapsto \max\{|x_i - x_{i-1}| \mid i = 1, 2, \dots, n\} \text{ where } P = \{x_0, x_1, \dots, x_n\}$$

**Theorem 0.1.13.** Let  $f : [a, b] \rightarrow \mathbb{R}$  and  $A \in \mathbb{R}$ , TFAE:

1.  $f$  is Riemann-Integral with  $\int_a^b f = A$ .
2. For any  $\varepsilon > 0$ , there exists  $\delta > 0$  such that  $\forall P \in \mathcal{P}[a, b], \|P\| < \delta \implies |R(f, P) - A| < \varepsilon$ .
3. For any  $\varepsilon > 0$ , there exists  $\delta > 0$  such that  $\forall P \in \mathcal{P}[a, b], P \supseteq P_0 \implies |R(f, P) - A| < \varepsilon$ .

The statement 2) can be written as:  $\lim_{\|P\| \rightarrow 0} R(f, P) = A$ .

Proof of Thm 0.1.13.

Enough to show  $1) \iff 2)$ . Let  $\int_a^b f = A$ .

Given  $\varepsilon > 0$ , there exists  $P_0 = \{x_0, x_1, \dots, x_n\}$  s.t.  $U(f, P_0) < A + \varepsilon$ .

Put  $M = \sup |f|$ ,  $\delta_1 = \frac{\varepsilon}{nM}$ . Let  $P \in \mathcal{P}[a, b]$  s.t.  $\|P\| < \delta_1$ . Write  $P = \{y_0, \dots, y_m\}$ .

Let  $I = \{i \in \{1, \dots, m\} \mid (y_{i-1}, y_i) \cap P_0 \neq \emptyset\}$  ( $M \geq 0, f > 0$ . ( $f = f + \min f$ ))

$$J = \{j \in \{1, \dots, m\} \mid (y_{j-1}, y_j) \cap P_0 = \emptyset\}$$

$$\begin{aligned} \text{Now, } U(f, P) &= \sum_{i=1}^m M_i (y_i - y_{i-1}) \\ &= \sum_{i \in I} M_i (y_i - y_{i-1}) + \sum_{i \in J} M_i (y_i - y_{i-1}) \\ &< nM \|P\| + U(f, P_0) < A + 2\varepsilon. \end{aligned}$$

점의 개수 많아서  $n$ 개! 더 큰 구간에서  $\sup$ !

Similarly  $L(f, P) > A - 2\varepsilon$ , for some  $\delta_2 > 0$ ,  $\|P\| < \delta_2$

**Theorem 0.1.14.** Let  $f : [a, b] \rightarrow \mathbb{R}$  be a Riemann-Integrable function. Define

$$F : [a, b] \rightarrow \mathbb{R} : x \mapsto \int_a^x f(t)dt$$

Then,  $F$  is Continuous on  $[a, b]$ . Particular, if  $f$  is continuous at  $x_0 \in [a, b]$ , then  $F$  is differentiable at  $x_0$  with

$$F'(x_0) = f(x_0)$$

**Theorem 0.1.15. The fundamental theorem of calculus**

If  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann-Integrable and there exists differentiable function  $F : [a, b] \rightarrow \mathbb{R}$  such that  $F' = f$ , then

$$\int_a^b f(x)dx = F(b) - F(a)$$

### 0.1.4 Bounded Variation Function

**Definition 0.1.16.** Let  $f: [a, b] \rightarrow \mathbb{R}$  be a bounded function and  $P = \{x_0, x_1, \dots, x_n\} \in \mathcal{P}[a, b]$ . Define a Variation of  $f$  with respects to  $P$  as:

$$V_a^b(f, P) \stackrel{\text{def}}{=} \sum_{i=1}^n |f(x_i) - f(x_{i-1})|$$

If the set  $\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$  is bounded above, then  $f$  is called *Bounded Variation Function*. In this case, define a *Total Variation* of  $f$  as:

$$TV_a^b(f) \stackrel{\text{def}}{=} \sup\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$$

In simply, denote  $V(f, P) = V_a^b(f, P)$  and  $TV(f) = TV_a^b(f)$ .

**Lemma 0.1.17.** Let  $f, g: [a, b] \rightarrow \mathbb{R}$  be Bounded Variation Functions,  $\alpha \in \mathbb{R}$  be a real number. Then,  $f + g, fg, \alpha f$  are Bounded Variation Functions and satisfy

$$\begin{aligned} TV(f + g) &\leq TV(f) + TV(g) \\ TV(fg) &\leq \sup_{t \in [a, b]} |g(t)| \cdot TV(f) + \sup_{t \in [a, b]} |f(t)| \cdot TV(g) \\ TV(\alpha f) &= |\alpha| \cdot TV(f) \end{aligned}$$

**Lemma 0.1.18.** Let  $f: [a, b] \rightarrow \mathbb{R}$  be a map, and  $a < c < b$ .

$f$  is bounded variation on  $[a, b]$  if and only if  $f$  is bounded variation on  $[a, c]$  and  $[c, b]$

In this case, the following equality holds:

$$TV_a^c(f) + TV_c^b(f) = TV_a^b(f)$$

**Theorem 0.1.19.** Let  $f: [a, b] \rightarrow \mathbb{R}$  be given.

$f$  is bouned variation if and only if For some monotone increasing  $g, h: [a, b] \rightarrow \mathbb{R}$ ,  $f = g - h$

Proof. Suppose  $f$  is bounded variation. Define

$$g(x) = V_a^x(f) \quad \text{and} \quad h(x) = V_a^x(f) - f(x).$$

By definition,  $g$  is monotone increasing. And, if  $a \leq x < y \leq b$ , then

$$h(y) - h(x) = V_a^y(f) - V_a^x(f) - (f(y) - f(x)) = V_x^y(f) - (f(y) - f(x)) \geq 0.$$

The converse is trivial.  $\square$

**Proposition 0.1.20.** Let  $f : [a, b] \rightarrow \mathbb{R}$  be a continuously differentiable function. If  $f$  is bounded variation, then

$$\int_a^b |f'(x)| dx \leq TV_a^b(f)$$

*Proof.* Let  $P = \{x_0, \dots, x_n\}$  be a Partition on  $[a, b]$ . By the Mean-Value Theorem, for each  $1 \leq i \leq n$ , for some  $t_i \in [x_{i-1}, x_i]$ ,

$$V_a^b(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i$$

Meanwhile,

$$L(|f'|, P) \leq \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq U(|f'|, P)$$

Both Lower Sum and Upper Sum converge to the  $\int_a^b |f'(x)| dx$ , we can complete this conversation.

Given  $\varepsilon > 0$ , there exists a partition  $P$  on  $[a, b]$  such that

$$\int_a^b |f'(x)| dx \leq V_a^b(f, P) + \varepsilon \leq TV_a^b(f) + \varepsilon$$

Since  $\varepsilon > 0$  was arbitrary, therefore we obtain the result. □

Note: if  $f$  is continuously differentiable, then

For any  $\varepsilon > 0$ , there exists a partition  $P$  on  $[a, b]$  such that  $\left| \int_a^b |f'(x)| dx - V_a^b(f, P) \right| < \varepsilon$ .

And, we can weaken the condition: continuously differentiable  $\rightarrow$  differentiable and the derivative is Riemann-Integrable.

**Proposition 0.1.21.** If  $f : [a, b] \rightarrow \mathbb{R}$  is differentiable and the derivative  $f'$  is bounded, then  $f$  is bounded variation.

*Proof.* For any partition  $P$  on  $[a, b]$ , by the Mean Value Theorem, there exists  $t_i \in (x_{i-1}, x_i)$  such that

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n \sup |f'| \cdot \Delta x_i = \sup |f'| \cdot 1$$

Since  $P$  was arbitrary,  $f$  is bounded variation. □

**Definition 0.1.22.** Let  $\alpha : [a, b] \rightarrow \mathbb{R}$  be a bounded variation.

A map  $f$  is said to be *Riemann-Stieltjes integrable* with respect to  $\alpha$  if:

for some monotone increasing  $\alpha_1, \alpha_2$ ,  $\alpha = \alpha_1 - \alpha_2$  such that  $f \in \mathcal{R}(\alpha_1)$  and  $f \in \mathcal{R}(\alpha_2)$

and in this case, define

$$\int_a^b f d\alpha = \int_a^b f d\alpha_1 - \int_a^b f d\alpha_2$$

**Lemma 0.1.23.** If  $f : [a, b] \rightarrow \mathbb{R}$  is continuous, then  $f \in \mathcal{R}(\alpha)$  for any bounded variation  $\alpha$ .

**Theorem 0.1.24.** Let  $f, \alpha : [a, b] \rightarrow \mathbb{R}$  be bounded variation functions. If  $f \in \mathcal{R}(\alpha)$ , then  $\alpha \in \mathcal{R}(f)$  with

$$\int_a^b f d\alpha + \int_a^b \alpha df = f(b)\alpha(b) - f(a)\alpha(a)$$

**Corollary 0.1.25.** If  $f : [a, b] \rightarrow \mathbb{R}$  is monotone increasing and  $g : [a, b] \rightarrow \mathbb{R}$  is continuous. Then, there exists  $c \in [a, b]$  such that

$$\int_a^b f(x)g(x)dx = f(a) \int_a^c g(x)dx + f(b) \int_c^b g(x)dx$$

### 0.1.5 Step function

**Definition 0.1.26.** Define a *Step function* as:

$$I: \mathbb{R} \rightarrow \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}$$

**Lemma 0.1.27.** If  $a < s < b$ ,  $f$  is bounded map on  $[a, b]$ ,  $f$  is continuous at  $s$ , and  $\alpha(x) = I(x - s)$ , then

$$\int_a^b f d\alpha = f(s)$$

**Theorem 0.1.28.** Suppose that  $c_n \geq 0$  and  $\sum c_n$  converges. And,  $\{s_n\}$  is a sequence of distinct points in  $(a, b)$ . Define

$$\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$$

If  $f$  is Continuous on  $[a, b]$ , then

$$\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$$